

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	19 June 2026	Team ID	24EG105H34
Project Name	HouseHunt – House Rent Management System	Maximum Marks	4 Marks

Technical Architecture

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web application interface for tenants, owners and admin	React.js, HTML, CSS, JavaScript, Bootstrap
2	Application Logic-1	User Authentication and Authorization	Node.js, Express.js, JWT, bcryptjs
3	Application Logic-2	Property Management Operations	Node.js, Express.js
4	Application Logic-3	Booking Management and Search Operations	Node.js, Express.js
5	Database	Stores users, properties and booking data	MongoDB
6	Cloud Database	Not Applicable (Local Database)	MongoDB Atlas (Future Scope)
7	File Storage	Property Images and Application Files	Local File System
8	External API-1	REST API Communication between Frontend and Backend	Axios
9	External API-2	Not Applicable	N/A
10	Machine Learning Model	Not Applicable	N/A
11	Infrastructure (Server / Cloud)	Application Deployment Environment	Local Server, VS Code, Node.js Runtime



Application Characteristics

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used for frontend, backend and database development	React.js, Node.js, Express.js, MongoDB, Bootstrap
2	Security Implementations	User authentication, password encryption and access control	JWT Authentication, bcryptjs Encryption, Role-Based Access Control
3	Scalable Architecture	Three-tier architecture supporting future enhancements	React.js Frontend, Express.js Backend, MongoDB Database
4	Availability	Accessible through web browsers anytime	Web-Based Application
5	Performance	Fast response and efficient database operation.	MongoDB Indexing, REST APIs, Axios

Technology Stack Overview

 **MongoDB**
NoSQL database for flexible data storage

 **Express.js**
Backend framework for REST APIs

 **React.js**
Frontend library for interactive UI

 **Node.js**
Server-side JavaScript runtime